

KUSHAGRA MISHRA

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SUMMARY

AI/ML engineer and 2026 Computer Science graduate, most recently an AI Engineering Intern at Tech Mahindra (Makers Lab), with hands-on experience building and deploying production agentic AI systems using LangGraph, Retrieval-Augmented Generation (RAG) pipelines, ChromaDB vector search, FastAPI, and React. Strong Java fundamentals and from-scratch ML experience in PyTorch. Available immediately for AI/ML Engineer or GenAI Developer roles.

PROFESSIONAL EXPERIENCE

AI Engineering Intern — Tech Mahindra, Makers Lab, Hyderabad

Jun 2026 – Aug 2026

- Built an in-app AI guidance chatbot — a 4-node LangGraph pipeline (intent routing, knowledge retrieval, capability matching, response generation) with FastAPI backend and React frontend
- Implemented semantic search using ChromaDB vector embeddings (all-MiniLM-L6-v2), replacing full-context injection with relevance-based retrieval for both knowledge base and capability matching
- Optimized pipeline from 3 to 2 large language model (LLM) calls per query (~33% fewer calls); deployed full stack to production (Render + ChromaDB Cloud + Netlify) with a documented GCP migration path (Cloud Run, Firebase Hosting, Secret Manager)
- Redesigned session management with server-side routing and session-scoped storage; authored technical reports on agentic architectures, vector databases, and GCP deployment

PROJECTS

MicroGPT + Attention X-Ray *Python, PyTorch, React*

microgpt-attention-xray.netlify.app

- Implemented a decoder-only transformer from scratch in PyTorch — multi-head causal self-attention, learned positional embeddings, pre-norm residual blocks, and a manual training loop on torch primitives only; trained 4 variants from 10.8M to 28.5M parameters (256-token context), reaching 1.49 validation loss vs a 2.46 bigram baseline
- Hand-wrote a byte-pair encoding tokenizer (3,000 merges, 3,065-token vocabulary) with no tokenizer library, training parallel character-level and subword models
- Built an attention visualization UI rendering all 64 post-softmax attention maps (8 layers × 8 heads) as hand-drawn SVG arcs, served by a 7-endpoint FastAPI backend with lazy, lock-guarded model caching to fit a 512MB deployment ceiling

Enterprise AI Workspace *React, FastAPI, MySQL, ChromaDB, LangChain*

enterprise-ai-workspace.netlify.app

- Cut API container memory below a 512MB deployment ceiling by removing the PyTorch/transformers/ONNX dependency chain and moving embedding generation to a hosted inference API; deployed across 6 managed services (Render, Aiven, Chroma Cloud, Hugging Face, Groq, S3)
- Built a multi-tenant RAG pipeline over per-user ChromaDB collections — recursive character chunking at 1,000 characters with 15% overlap, 384-dimension MiniLM embeddings, top-4 retrieval — with module-level client reuse to avoid per-query re-initialization

Autonomous Code Review Swarm *Python, LangGraph, MCP*

github.com/Kushagra-Mishra1008/review_swarm

- Built a hierarchical multi-agent pull-request review system — a lead orchestrator over 4 domain specialists (security, performance, testing, maintainability) fanning out to concurrent per-file worker agents, with pre-LLM activation rules that skip specialists a diff does not warrant; reviewed a 10-file PR in 39.8s across 9,897 tokens with zero rate-limit failures
- Engineered a centralized LLM gateway enforcing sliding-window token-bucket rate limiting, SHA-256-keyed disk caching, and 4-way task routing across two model tiers to hold the swarm inside an 8K tokens-per-minute budget
- Authored a custom Model Context Protocol (MCP) server exposing 3 code-search tools over AST-chunked repositories, fusing vector and exact-match results by reciprocal rank; integrated with the official GitHub and Filesystem MCP servers

SKILLS

Languages: Java, Python, JavaScript, SQL

AI & GenAI: LangChain, LangGraph, CrewAI, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Integration, PyTorch, Transformers

Vector & ML: ChromaDB, Vector Embeddings, Semantic Search, Sentence Transformers, Scikit-learn, Pandas, NumPy

Backend & Web: Spring Boot, FastAPI, REST APIs, React.js, HTML5, CSS3

DevOps & Cloud: Git, GitHub, Docker, Render, Netlify, GCP (Cloud Run, Firebase Hosting, Secret Manager), Postman

EDUCATION

B.Tech, Computer Science and Engineering — Vellore Institute of Technology Sep 2022 – May 2026 | CGPA: 8.19

Class XII (CBSE) — Jawahar Vidya Mandir, Ranchi

2022 | 89%

Class X — St. Thomas School, Ranchi

2020 | 96%